In [1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```python
%pip install mysql-connector-python
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: mysql-connector-python in c:\users\asus\appdata\roaming\python\python312\site-pac
kages (9.5.0)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```python
import mysql.connector

# Connect to server
cnx = mysql.connector.connect(
    host="127.0.0.1",
    port=3306,
    user="root",
    password="Password")
```

In [4]:
```python
query = "SELECT * FROM telecom_churn_project.churn_data"
df = pd.read_sql(query, cnx)
```

```
C:\Users\ASUS\AppData\Local\Temp\ipykernel_20668\3369739163.py:2: UserWarning: pandas only supports SQLAlchemy c
onnectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not
tested. Please consider using SQLAlchemy.
  df = pd.read_sql(query, cnx)
```

In [5]:
```python
df.head()
```

Out[5]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	..
0	7590-VHVEG	Female	0	Yes	No	1	No	No	DSL	No	..
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	..
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	..
3	7795-CFOCW	Male	0	No	No	45	No	No	DSL	Yes	..
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	..

5 rows × 21 columns

In [6]:
```python
df.shape
```

Out[6]:
```
(7032, 21)
```

In [7]:
```python
df.info
```

Out[7]:
```
<bound method DataFrame.info of       customerID  gender  SeniorCitizen Partner Dependents  tenure  \
0     7590-VHVEG  Female              0     Yes         No       1
1     5575-GNVDE    Male              0      No         No      34
2     3668-QPYBK    Male              0      No         No       2
3     7795-CFOCW    Male              0      No         No      45
4     9237-HQITU  Female              0      No         No       2
...          ...     ...            ...     ...        ...     ...
7027  6840-RESVB    Male              0     Yes        Yes      24
7028  2234-XADUH  Female              0     Yes        Yes      72
7029  4801-JZAZL  Female              0     Yes        Yes      11
7030  8361-LTMKD    Male              1     Yes         No       4
7031  3186-AJIEK    Male              0      No         No      66

     PhoneService MultipleLines InternetService OnlineSecurity  ...  \
0              No            No             DSL             No  ...
1             Yes            No             DSL            Yes  ...
2             Yes            No             DSL            Yes  ...
3              No            No             DSL            Yes  ...
4             Yes            No     Fiber optic             No  ...
...           ...           ...             ...            ...  ...
7027          Yes           Yes             DSL            Yes  ...
7028          Yes           Yes     Fiber optic             No  ...
7029           No            No             DSL            Yes  ...
7030          Yes           Yes     Fiber optic             No  ...
7031          Yes            No     Fiber optic            Yes  ...

     DeviceProtection TechSupport StreamingTV StreamingMovies        Contract  \
0                  No          No          No              No  Month-to-month
1                 Yes          No          No              No        One year
2                  No          No          No              No  Month-to-month
3                 Yes         Yes          No              No        One year
4                  No          No          No              No  Month-to-month
...               ...         ...         ...             ...             ...
7027              Yes         Yes         Yes             Yes        One year
7028              Yes          No         Yes             Yes        One year
7029               No          No          No              No  Month-to-month
7030               No          No          No              No  Month-to-month
7031              Yes         Yes         Yes             Yes        Two year

     PaperlessBilling              PaymentMethod MonthlyCharges  TotalCharges  \
0                 Yes           Electronic check          29.85         29.85
1                  No               Mailed check          56.95       1889.50
2                 Yes               Mailed check          53.85        108.15
3                  No  Bank transfer (automatic)          42.30       1840.75
4                 Yes           Electronic check          70.70        151.65
...               ...                        ...            ...           ...
7027              Yes               Mailed check          84.80       1990.50
7028              Yes    Credit card (automatic)         103.20       7362.90
7029              Yes           Electronic check          29.60        346.45
7030              Yes               Mailed check          74.40        306.60
7031              Yes  Bank transfer (automatic)         105.65       6844.50

      Churn
0        No
1        No
2       Yes
3        No
4       Yes
...     ...
7027     No
7028     No
7029     No
7030    Yes
7031     No

[7032 rows x 21 columns]>
```

In [8]:
```
df.describe()
```

Out[8]:

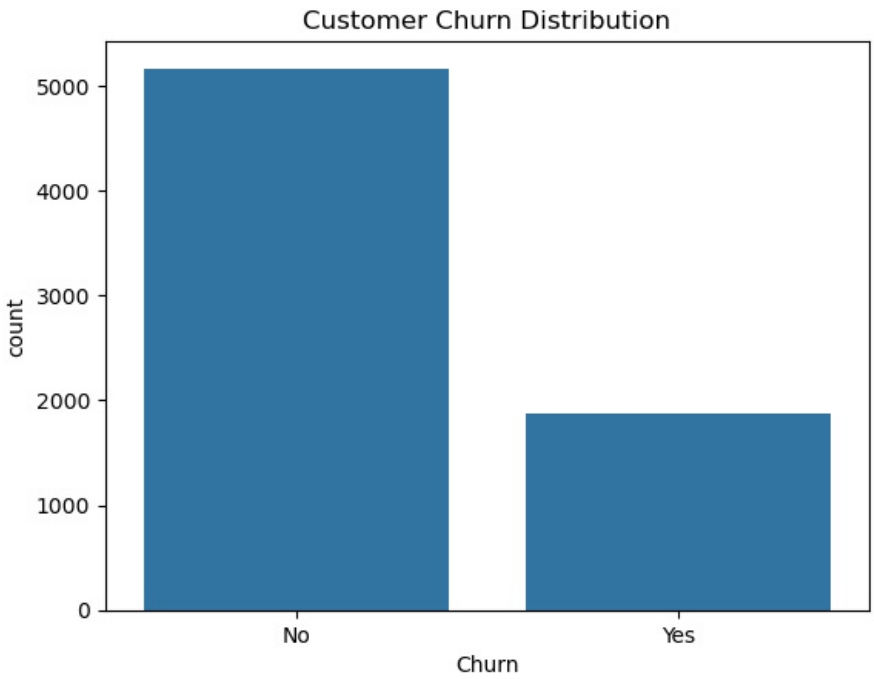
	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7032.000000	7032.000000	7032.000000	7032.000000
mean	0.162400	32.421786	64.798208	2283.300441
std	0.368844	24.545260	30.085974	2266.771362
min	0.000000	1.000000	18.250000	18.800000
25%	0.000000	9.000000	35.587500	401.450000
50%	0.000000	29.000000	70.350000	1397.475000
75%	0.000000	55.000000	89.862500	3794.737500
max	1.000000	72.000000	118.750000	8684.800000

In [9]:
```
df.columns
```

Out[9]:
```
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='object')
```

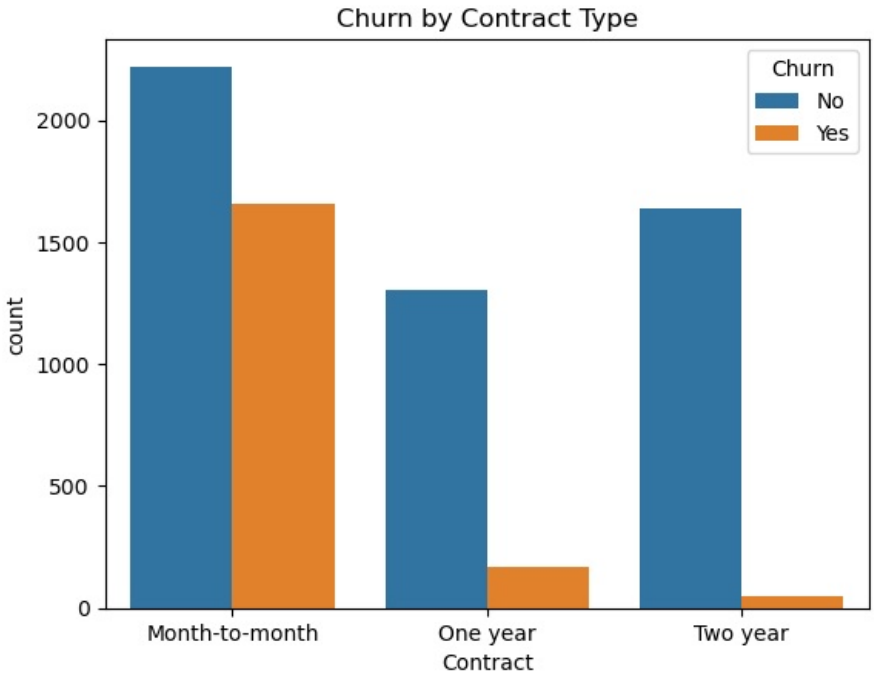
Churn Distribution

In [11]:
```
sns.countplot(x='Churn', data=df)
plt.title("Customer Churn Distribution")
plt.show()
```



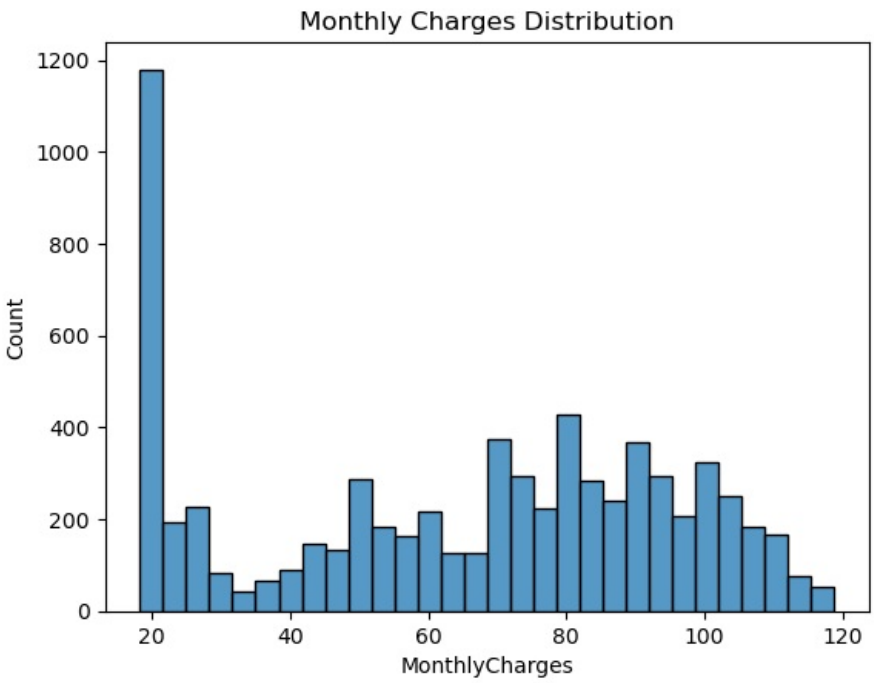
Churn by Contract Type

In [13]:
```
sns.countplot(x='Contract', hue='Churn', data=df)
plt.title("Churn by Contract Type")
plt.show()
```



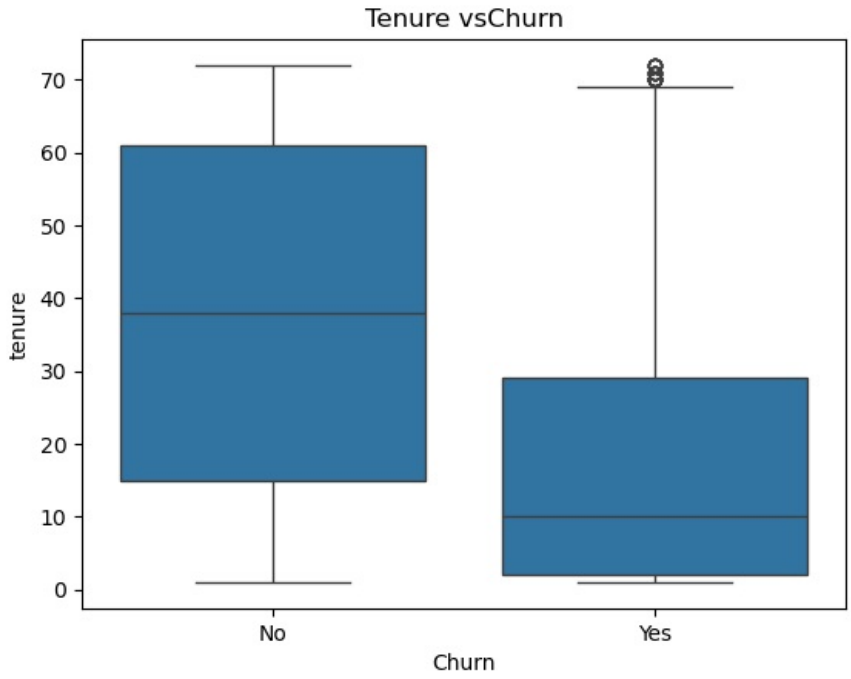
Monthly Charges Distribution

In [15]:
```
sns.histplot(df['MonthlyCharges'], bins=30)
plt.title("Monthly Charges Distribution")
plt.show()
```



Tenure vsChurn

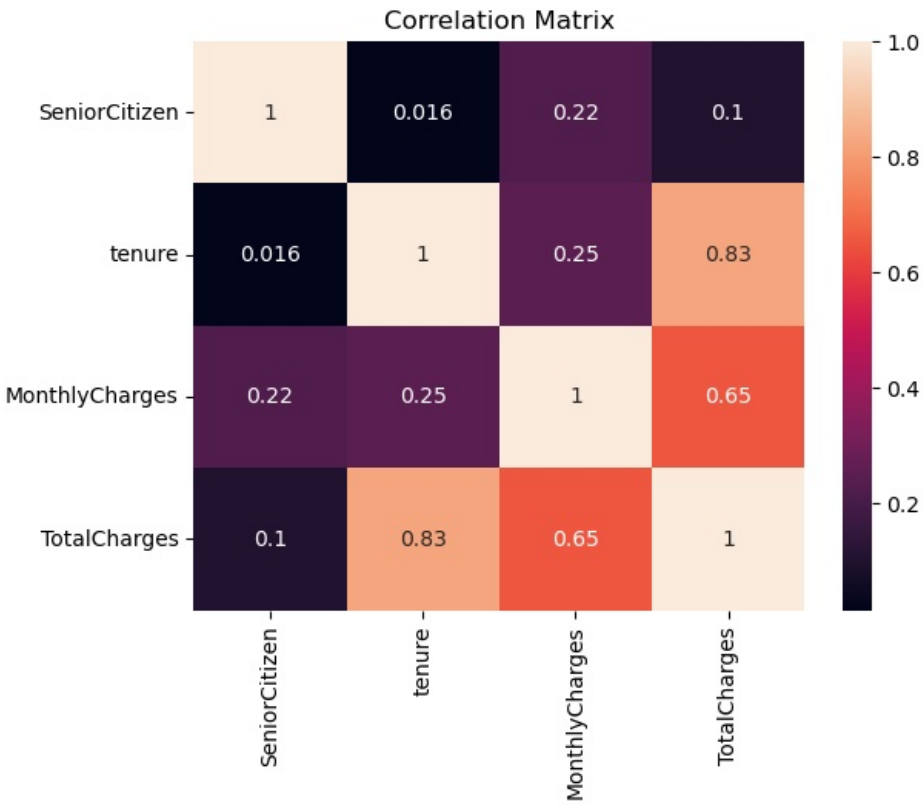
In [17]:
```
sns.boxplot(x='Churn', y='tenure', data=df)
plt.title("Tenure vsChurn")
plt.show()
```



Correlation Matrix

In [19]:
```
corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True)
plt.title("Correlation Matrix")
plt.show()
```



In [ ]:

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